

# Manish Motwani

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 School of Electrical Engineering & Computer Science  
Oregon State University  
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 <https://mmotwani.com/>

## Research Interests

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My broad research interests lie in improving software engineers' productivity by automating software engineering practices. I design novel automation techniques for tasks such as requirements elicitation, software testing, and program repair, targeting diverse application domains including artificial intelligence (AI), high-performance computing (HPC), network protocols, and REST API-based web applications.

## Science and Technology Specialties

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- **Software Testing & Verification:** automated test generation, test oracle synthesis, differential fuzzing, mutation testing, flaky-test detection, black-box and grey-box testing.
- **Automated Program Repair:** generate-and-validate repair, semantics-guided repair, fault localization, patch overfitting mitigation, LLM-assisted repair.
- **Program Analysis:** static analysis, dynamic analysis, symbolic execution, specification inference, REST API analysis.
- **AI & Machine Learning for Software Engineering:** large language model (LLM) testing, LLM-guided fuzzing, neural-network-based program repair, GNN-based fault localization.
- **High-Performance Computing (HPC) Software Quality:** heterogeneous-memory bug detection, GPU platform-specific bug detection, energy-aware testing, HPC code translation analysis.
- **Requirements Engineering:** natural-language processing for requirements elicitation, use-case and validation extraction, automated test oracle generation from structured specifications.
- **Security & Compliance Automation:** certificate authority testing, security compliance value chain automation, automated interpretation of legal regulations.

## Education

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UNIVERSITY OF MASSACHUSETTS AMHERST ..... Amherst, MA, USA

Sept 2015 – Sept 2022    **Doctor of Philosophy in Computer Science**  
Dissertation: High-Quality Automatic Program Repair  
Advisor: Prof. Yuriy Brun

Sept 2015 – May 2018    **Master of Science in Computer Science**  
Field of Study: Software Engineering, Automated Program Repair

INTERNATIONAL INSTITUTE OF INFORMATION TECHNOLOGY, HYDERABAD ..... Hyderabad, TS, India

Aug 2007 – May 2011    **Bachelor of Technology (Honors) in Computer Science and Engineering**

## Work Experience

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- SCHOOL OF ELECTRICAL ENGINEERING AND COMPUTER SCIENCE, OREGON STATE UNIVERSITY . . . . . Corvallis, OR, USA  
Sept 2023 – Present      **Assistant Professor (tenure-track)**  
Conduct independent research in software engineering (software testing, automated program repair, and program analysis targeting HPC, AI, and REST API-based systems). Design and teach graduate and undergraduate software engineering courses. Advise and mentor MS and PhD students. Develop and submit external research proposals to federal funding agencies (NSF, ARL, DOE). Perform departmental and professional service including faculty search committees, undergraduate research committees, and program committee memberships at top software engineering conferences.
- COLLEGE OF COMPUTING – SCHOOL OF COMPUTER SCIENCE, GEORGIA INSTITUTE OF TECHNOLOGY . . . Atlanta, GA, USA  
Aug 2022 – Aug 2023      **Postdoctoral Fellow**  
Conducted research on automated REST API specification generation using static program analysis (resulting in an ICSE 2024 publication) and on heterogeneous memory bug detection in high-performance computing applications. Mentored PhD and MS students. Collaborated with researchers at Georgia Tech and national labs.
- SECURITY TOOLS GROUP, ONE ENGINEERING SYSTEMS, MICROSOFT . . . . . Redmond, WA, USA  
Aug 2021 – Oct 2021      **Research Intern**  
Conducted research on creating automated software security and compliance tool within Microsoft's internal engineering systems organization.
- May 2020 – Aug 2020      **Research Intern**  
Conducted research on analyzing the automated security compliance tooling and process for large-scale enterprise codebases.
- MANNING COLLEGE OF INFORMATION AND COMPUTER SCIENCES, UNIVERSITY OF MASSACHUSETTS . Amherst, MA, USA  
Sept 2015 – July 2022      **Research Assistant**  
Conducted doctoral research on high-quality automated program repair, including fault localization, test oracle generation from natural-language specifications, and evaluation of APR techniques on real-world defects. Assisted in teaching CS 520 (Theory and Practice of Software Engineering) as a teaching assistant and guest lecturer. Mentored MS students on independent research projects.
- TATA RESEARCH DEVELOPMENT AND DESIGN CENTRE (TRDDC) . . . . . Pune, MH, India  
July 2011 – July 2015      **Researcher**  
Conducted applied research in requirements engineering, natural language processing for software artifacts, and ontology-based knowledge representation. Mentored junior researchers and BS students. This position falls partially outside the mandatory 10-year window but is included for completeness.
- CISCO SYSTEMS INDIA PVT. LTD. . . . . Bangalore, KA, India  
May 2010 – July 2010      **Software Developer Intern**  
Developed software components for router as part of a summer internship with Cisco's India engineering team. This position falls outside the mandatory 10-year window but is included for completeness.

## Teaching

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### Teaching Philosophy and Approach

My teaching philosophy is based on the belief that software engineering is fundamentally collaborative and systems oriented. Students learn best by building substantial systems, working in teams, engaging with real users, and iterating through authentic engineering practices. My teaching emphasizes real-world relevance, modern paradigms (AI-based development), and collaborative problem-solving. I am committed to developing and sharing practical models that help other faculty create similarly authentic learning experiences in their own courses.

### Teaching Experience

SCHOOL OF ELECTRICAL ENGINEERING AND COMPUTER SCIENCE, OREGON STATE UNIVERSITY . . . . . Corvallis, OR, USA

|             |                                                                                                       |
|-------------|-------------------------------------------------------------------------------------------------------|
| Spring 2026 | Instructor (CS 569: <i>Selected Topics in Software Engineering: Program Analysis and Evaluation</i> ) |
| Winter 2026 | Instructor (CS 563: <i>Software Evolution and Maintenance</i> )                                       |
| Winter 2026 | Instructor (CS 362: <i>Software Engineering II</i> )                                                  |
| Winter 2025 | Instructor (CS 563: <i>Software Evolution and Maintenance</i> )                                       |
| Winter 2025 | Instructor (CS 362: <i>Software Engineering II</i> )                                                  |
| Spring 2024 | Instructor (CS 563: <i>Software Evolution and Maintenance</i> )                                       |
| Winter 2024 | Instructor (CS 569: <i>Selected Topics in Software Engineering: Program Analysis and Evaluation</i> ) |

MANNING COLLEGE OF INFORMATION AND COMPUTER SCIENCES, UNIVERSITY OF MASSACHUSETTS . Amherst, MA, USA

|             |                                                                                  |
|-------------|----------------------------------------------------------------------------------|
| Fall 2021   | Guest Lecturer (CS 520: <i>Theory and Practice of Software Engineering</i> )     |
| Spring 2020 | Guest Lecturer (CS 520: <i>Theory and Practice of Software Engineering</i> )     |
| Fall 2018   | Teaching Assistant (CS 520: <i>Theory and Practice of Software Engineering</i> ) |

TATA RESEARCH DEVELOPMENT AND DESIGN CENTRE . . . . . Pune, MH, India

|          |                                                                                              |
|----------|----------------------------------------------------------------------------------------------|
| May 2013 | Guest Lecturer ( <i>Modelling Business Processes Using the Web Ontology Language (OWL)</i> ) |
|----------|----------------------------------------------------------------------------------------------|

INTERNATIONAL INSTITUTE OF INFORMATION TECHNOLOGY, HYDERABAD . . . . . Hyderabad, TS, India

|             |                                                                 |
|-------------|-----------------------------------------------------------------|
| Spring 2011 | Teaching Assistant (CEG421: <i>Building Energy Simulation</i> ) |
| Fall 2010   | Teaching Assistant (CS3155: <i>Compilers</i> )                  |
| Fall 2009   | Teaching Assistant (MA3200: <i>Discrete Maths</i> )             |

## Publications

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### Referred Journal/Conference Publications

- Rahil Mehta, Pushpak Katkhede, and **Manish Motwani**, Defects4REST: A Benchmark of Real-World Defects to Enable Controlled Testing and Debugging Studies for REST APIs. In *proceedings of the 48th ACM/IEEE International Conference on Software Engineering (ICSE)*, Rio de Janeiro, Brazil, April 2026. DOI: 10.1145/3744916.3787787
- **Manish Motwani**, Aakash Kulkarni, Yunhan Qiao, Matthew Davis, and Ziyang Chen, LLM-Guided Differential Fuzzing for Detecting Platform-Specific Bugs in Scientific Applications. In *proceedings of the IEEE International Conference on AI x Software Engineering (AIxSE)*, pages 59–66, Laguna Hills, CA, USA, October 2025. DOI: 10.1109/AIxSE64906.2025.00015
- Ruikai Huang, **Manish Motwani**, Idel Martinez, and Alessandro Orso, Generating REST API Specifications through Static Analysis. In *proceedings of the research track at the 46th International Conference on Software Engineering (ICSE)*, Lisbon, Portugal, April 2024. ACM artifact badges granted: Artifact Available, Artifact Reusable. DOI: 10.1145/3597503.3639137
- **Manish Motwani** and Yuriy Brun, Better Automatic Program Repair by Using Bug Reports and Tests Together. In *proceedings of the technical track at the 45<sup>th</sup> International Conference on Software Engineering (ICSE)*, pages 1225–1237, Melbourne, Australia, May 2023. ACM artifact badges granted: Artifact Available, Artifact Reusable.

- **Manish Motwani** and Yuriy Brun, Understanding Why and Predicting When Developers Adhere to Code-Quality Standards. In *proceedings of the Software Engineering in Practice track at the 45<sup>th</sup> International Conference on Software Engineering (ICSE-SEIP)*, pages 432–444, Melbourne, Australia, May 2023. DOI: 10.1109/ICSE-SEIP58684.2023.00045
- **Manish Motwani**, Mauricio Soto, Yuriy Brun, René Just, and Claire Le Goues, Quality of Automated Program Repair on Real-World Defects, *IEEE Transactions on Software Engineering (TSE)*, 2020. DOI: 10.1109/TSE.2020.2998785
- **Manish Motwani** and Yuriy Brun, Automatically Generating Precise Oracles from Structured Natural Language Specifications. In *proceedings of the technical track at the 41<sup>st</sup> International Conference on Software Engineering (ICSE)*, pages 188–199, Montreal, QC, Canada, May 2019. ACM artifact badges granted: Artifact Available, Artifact Reusable. DOI: 10.1109/ICSE.2019.00035
- Afsoon Afzal, **Manish Motwani**, Kathryn T. Stolee, Yuriy Brun, and Claire Le Goues, SOSRepair: Expressive Semantic Search for Real-World Program Repair, *IEEE Transactions on Software Engineering (TSE)*, 2019. DOI: 10.1109/TSE.2019.2944914
- **Manish Motwani**, Sandhya Sankaranarayanan, René Just, and Yuriy Brun, Do Automated Program Repair Techniques Repair Hard and Important Bugs?, In *proceedings of the Journal First track at the 40<sup>th</sup> International Conference on Software Engineering (ICSE)*, pages 25, Gothenburg, Sweden, May 2018.
- **Manish Motwani**, Sandhya Sankaranarayanan, René Just, and Yuriy Brun, Do Automated Program Repair Techniques Repair Hard and Important Bugs?, *Empirical Software Engineering (EMSE)*, 2018. DOI: 10.1007/s10664-017-9550-0

### Referred Short Conference Publications

- Ninad Anklesaria and **Manish Motwani**, Energy-Aware Test Prioritization for High-Performance Computing: A Multi-Objective Approach. In *companion proceedings of the ACM International Conference on the Foundations of Software Engineering (FSE)*, June 5–9, 2026, Montreal, Canada.
- **Manish Motwani**, High-Quality Automated Program Repair. In *companion proceedings of the IEEE/ACM 43<sup>rd</sup> International Conference on Software Engineering (ICSE-Companion)*, pages 309–314, Virtual (originally in Madrid, Spain), May 2021. DOI: 10.1109/ICSE-Companion52605.2021.00134.
- Smita Ghaisas, **Manish Motwani**, Balaji Balasubramaniam, Anjali Gajendragadkar, Rahul Kelkar, and Harrick Vin. Towards Automating the Security Compliance Value Chain. In *proceedings of the Industrial Track at the 10<sup>th</sup> Joint Meeting on Foundations of Software Engineering (FSE)*, pages 1014–1017, Bergamo, Italy, August 2015. DOI: 10.1145/2786805.2804435
- Smita Ghaisas, **Manish Motwani**, and Preethu Rose. Detecting System Use Cases and Validations from Documents. In *proceedings of the New Ideas and Emerging Results Track at the 28<sup>th</sup> IEEE/ACM International Conference on Automated Software Engineering (ASE)*, pages 568–573, Palo Alto, CA, USA, November 2013. DOI: 10.1109/ASE.2013.6693114

### Dissertation

- **Manish Motwani**. High-Quality Automatic Program Repair. 2022. Doctoral Dissertations. 2696. University of Massachusetts, Amherst MA USA. DOI: 10.7275/30288519

### Referred Workshop Publications

- Yashwanthi Anand, Rahil P. Mehta, **Manish Motwani**, and Sandhya Saisubramanian, Uncovering Systemic and Environment Errors in Autonomous Systems Using Differential Testing. In *proceedings of the AAAI Fall Symposium Series on AI Trustworthiness and Risk Assessment for Challenged Contexts (AAAI 2025 FS – ATRACC)*, Arlington, VA, USA, Nov 2025.

- Preethu R. Anish, SK. Sharma, **Manish Motwani**, and Smita Ghaisas. Knowledge-assisted Product Requirements Configurator. In *proceedings of the 4<sup>th</sup> International Workshop on Product Line Approaches in Software Engineering (PLEASE)*, pages 29–32, San Francisco, CA, USA, May 2013. DOI: 10.1109/PLEASE.2013.6608660

## arXiv Preprints

- Matthew Davis, Aakash Kulkarni, Ziyang Chen, Yunhan Qiao, Christopher Terrazas, and **Manish Motwani**, Automatically Detecting Heterogeneous Bugs in High-Performance Computing Scientific Software. arXiv preprint arXiv:2501.09872, 2025.
- Rahil P. Mehta, Yashwanthi Anand, **Manish Motwani**, and Sandhya Saisubramanian, Uncovering Systemic and Environment Errors in Autonomous Systems Using Differential Testing. arXiv preprint arXiv:2507.03870, 2025.
- **Manish Motwani** and Yuriy Brun, Automatically Repairing Programs Using Both Tests and Bug Reports. arXiv preprint arXiv:2011.08340, 2020.

## Patents

- Smita Ghaisas, **Manish Motwani**, Preethu R. Anish, Balaji Balasubramaniam, and Aarth Krishnamurthy. Systems and Methods for an Automated Interpretation of Legal Regulations. United States Patent # US9972016B2, May 2018.
- Smita Ghaisas, **Manish Motwani**, Preethu Rose Anish, and Shashi Kant Sharma. Automated Classification of Business Rules from Text. United States Patent # US10146762B2, Dec 2018.

## Invited Talks and Panels

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- **Invited Talk**, Advanced Computing Systems Research, *Can We Trust Scientific Software Ported by Humans and AI? Detecting Porting Errors and Reducing Validation Costs*, Oak Ridge National Laboratory, June 2026.
- **Invited Talk**, AI Seminar, *AI for Software Engineering (AI4SE): Opportunities and Challenges*, Oregon State University, January 2024.
- **Panelist**, Birds of a Feather (BoF) Session on Software Testing for Scientific Computing in HPC, Supercomputing Conference (SC), Denver, USA, November 2023.
- **Panelist**, Key Challenges and Promising Research Directions for Program Repair, 4th International Workshop on Automated Program Repair (APR), Melbourne, Australia, May 2023.

## Formal Presentations

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- **LLM-Guided Differential Fuzzing for Detecting Platform-Specific Bugs in Scientific Applications.** The *Regular Paper Track of the IEEE International Conference on AI x Software Engineering (AIxSE)*, Laguna Hills, CA, USA, October 2025.
- **Better Automatic Program Repair by Using Bug Reports and Tests Together.** The *Technical Track of the 45<sup>th</sup> International Conference on Software Engineering (ICSE)*, Melbourne, Australia, May 2023.
- **Understanding Why and Predicting When Developers Adhere to the Code Quality Standards.** The *Software Engineering in Practice (SEIP) Track of the 45<sup>th</sup> International Conference on Software Engineering (ICSE)*, Melbourne, Australia, May 2023.
- **High Quality Automated Program Repair.** The *Doctoral Symposium Track of the 43<sup>rd</sup> International Conference on Software Engineering (ICSE)*, Virtual (originally Madrid, Spain), May 2021.
- **Automatically Generating Precise Oracles from Structured Natural Language Specifications.** The *Technical Track of the 41<sup>st</sup> International Conference on Software Engineering (ICSE)*, Montreal, QC, Canada, May 2019.
- **Do Automated Program Repair Techniques Repair Hard and Important Bugs?** The *Journal First Track of the*

40<sup>th</sup> International Conference on Software Engineering (ICSE), Gothenburg, Sweden, May 2018.

- **Detecting System Use Cases and Validations from Documents.** *The New Ideas and Emerging Results Track of the 28<sup>th</sup> IEEE/ACM International Conference on Automated Software Engineering (ASE)*, Palo Alto, CA, USA, November 2013.

## Research Mentoring

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- Akshith Gunasekaran (PhD in CS Graduate, Oregon State)  
**Automated Testing for Certificate Authority Software**, April 2024 – January 2026.
- Rahil Mehta (MS in CS graduate, Oregon State)  
**Automated Testing for REST API services**, Jan 2024 – June 2026.
- Aakash Kulkarni (MEng in CS graduate, Oregon State)  
**Automated Testing for High-Performance Computing Scientific Software**, April 2024 – June 2025.
- Kausar Moshood (1<sup>st</sup> year PhD in CS, Oregon State)  
**Automated Testing and Debugging of REST APIs**, Sept 2025 – Present.
- Ninad Anklesaria (1<sup>st</sup> year PhD in CS, Oregon State)  
**Energy-Aware Testing of High-Performance Computing Scientific Software**, Sept 2025 – Present.
- Oscar Ludwig (4<sup>th</sup> year Undergrad in CS, Oregon State)  
**Automated Testing for High-Performance Computing Scientific Software**, April 2025 – Present.
- Farzaan Wadiwalla (3<sup>rd</sup> year Undergrad in CS, Oregon State)  
**LLM-Guided Verification to Solve the Patch Overfitting Problem**, June 2025 – Present.
- Matthew Davis (2<sup>nd</sup> year PhD in CS, Georgia Tech)  
**Heterogeneous Memory Bug Detection in High-Performance Computing Applications**, Jan 2024 – June 2025.
- Ruikai Huang (3<sup>rd</sup> year PhD in CS, Georgia Tech)  
**Generating REST API Specifications through Static Analysis**, Sept 2022 – Aug 2023.
- Jerry Chen (1<sup>st</sup> year MS in CS, Georgia Tech)  
**Heterogeneous Memory Bug Detection in High-Performance Computing Applications**, Jan 2023 – Aug 2023.
- Sandhya Sankaranarayanan (1<sup>st</sup> year MS in CS, UMass Amherst)  
**Do Automated Program Repair Techniques Repair Hard and Important Bugs?**, Sept 2015 – May 2016.
- Priya Wagh, Madhuri Jadhav, and Nidhi Zanwar (1<sup>st</sup> year BS students in CS, MIT Pune)  
**Purpose-Centric Search for Enterprise Knowledge Reuse**, Aug 2012 – April 2013.

## Technical Skills

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- **Research & Planning:** Identifying Problems, Gathering Information, Developing Evaluations, Calculating Results
- **Programming Languages:** Working knowledge of Java, C, C++, C#, Python, Shell, R, Ruby, and PHP
- **Machine Learning Frameworks:** PyTorch, Keras/TensorFlow
- **Databases:** MySQL, Oracle, Kusto
- **Integrated Development Environments and Tools:** Eclipse, NetBeans, MS Visual Studio, Vim, Git, Mercurial
- **Cloud Platforms:** Amazon Web Services, SLURM clusters

## Open-Source Software and Software Artifacts

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- **Defects4REST:** The first executable benchmark of real-world defects in popular REST API services.  
<https://github.com/ANSWER-OSU/Defects4REST>

- **CAFuzz**: Black-box test generator that models RFC-defined workflows to systematically test certificate issuance, validation, revocation, and error handling in Certificate Authority implementations.  
<https://github.com/ANSWER-OSU/cafuzz>
- **HeteroBugDetect**: Automatically detecting heterogeneous bugs in HPC scientific applications.  
<https://doi.org/10.7910/DVN/VH09RH>
- **AIProbe**: Black-box testing framework to identify and attribute execution anomalies in autonomous AI agents.  
<https://github.com/ANSWER-OSU/AIProbe>
- **Respector**: Automatically generate REST API specifications in the OpenAPI format using static program analysis.  
<https://github.com/nntzuekai/Respector>
- **Blues**: Information retrieval-based statement-level fault localization for automated program repair.  
<https://github.com/LASER-UMASS/Blues>
- **RAFL**: Combine fault localization results of multiple techniques using unsupervised rank aggregation algorithms.  
<https://github.com/LASER-UMASS/RAFL>
- **SBIR replication package**: Replication package for evaluating SBIR (SBFL and Blues combined using RAFL) and the repair performance of three APR techniques using SBIR on the Defects4J benchmark.  
<https://github.com/LASER-UMASS/SBIR-ReplicationPackage>
- **JaRFly**: Java repair framework.  
<http://JaRFly.cs.umass.edu/>
- **JaRFly replication package**: Replication package of evaluating JaRFly on the Defects4J benchmark.  
<https://github.com/LASER-UMASS/JavaRepair-replication-package>
- **Swami**: Automated test generation from natural language software specifications.  
<https://Swami.cs.umass.edu>
- **SOSRepair**: Expressive semantic search for real-world program repair.  
<https://github.com/squaresLab/SOSRepair/>
- **SOSRepair replication package**: Replication package for evaluating SOSRepair on the ManyBugs benchmark.  
<https://github.com/squaresLab/SOSRepair-Replication-Package>
- **Repair Applicability**: Data, scripts, and methodology for evaluating the applicability of automated program repair.  
<https://github.com/LASER-UMASS/AutomatedRepairApplicabilityData/>

## Service

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### Professional Service

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| ..... 2026 .....   |                                                                                                                                       |
| PC member          | New Ideas and Emerging Results (NIER) track, 48 <sup>th</sup> IEEE/ACM International Conference on Software Engineering (ICSE)        |
| ..... 2025 .....   |                                                                                                                                       |
| PC member          | Research track, 34 <sup>th</sup> ACM SIGSOFT International Symposium on Software Testing and Analysis (ISSTA)                         |
| PC member          | Industry Government and Communities track, 19 <sup>th</sup> ACM/IEEE Empirical Software Engineering and Measurement (ESEM) conference |
| Reviewer           | IEEE Transactions on Software Engineering (TSE) journal                                                                               |
| Reviewer           | ACM Transactions on Software Engineering and Methodology (TOSEM) journal                                                              |
| Undergraduate      | Academic Year 2024–25, Oregon State University                                                                                        |
| Research Committee |                                                                                                                                       |

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|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| ..... 2024 .....                |                                                                                                                                         |
| PC member                       | Research track, 32 <sup>nd</sup> ACM International Conference on the Foundations of Software Engineering (FSE)                          |
| PC member                       | Industry Government and Communities track, 18 <sup>th</sup> ACM/IEEE Empirical Software Engineering and Measurement (ESEM) conference   |
| Reviewer                        | IEEE Transactions on Software Engineering (TSE) journal                                                                                 |
| Reviewer                        | ACM Transactions on Software Engineering and Methodology (TOSEM) journal                                                                |
| Faculty search committee member | Computer Science, Academic Year 2023–24, Oregon State University                                                                        |
| ..... 2023 .....                |                                                                                                                                         |
| Reviewer                        | IEEE Transactions on Software Engineering (TSE) journal                                                                                 |
| Sub-reviewer                    | 32 <sup>nd</sup> ACM SIGSOFT International Symposium on Software Testing and Analysis (ISSTA)                                           |
| ..... 2022 .....                |                                                                                                                                         |
| PC member                       | Challenge track at the 13 <sup>th</sup> Symposium on Search-Based Software Engineering (SSBSE)                                          |
| Reviewer                        | IEEE Transactions on Software Engineering (TSE) journal                                                                                 |
| Reviewer                        | ACM Transactions on Software Engineering and Methodology (TOSEM) journal                                                                |
| Sub-reviewer                    | 43 <sup>rd</sup> ACM/IEEE International Conference on Software Engineering (ICSE)                                                       |
| ..... 2021 .....                |                                                                                                                                         |
| PC member                       | Challenge track at the 13 <sup>th</sup> Symposium on Search-Based Software Engineering (SSBSE)                                          |
| Reviewer                        | IEEE Transactions on Software Engineering (TSE) journal                                                                                 |
| Reviewer                        | ACM Transactions on Software Engineering and Methodology (TOSEM) journal                                                                |
| ..... 2020 .....                |                                                                                                                                         |
| Reviewer                        | IEEE Transactions on Software Engineering (TSE) journal                                                                                 |
| Sub-reviewer                    | 42 <sup>nd</sup> ACM/IEEE International Conference on Software Engineering (ICSE)                                                       |
| Sub-reviewer                    | 35 <sup>th</sup> IEEE/ACM International Conference on Automated Software Engineering (ASE)                                              |
| ..... 2019 .....                |                                                                                                                                         |
| Reviewer                        | IEEE Transactions on Software Engineering (TSE) journal                                                                                 |
| Sub-reviewer                    | 41 <sup>st</sup> ACM/IEEE International Conference on Software Engineering (ICSE)                                                       |
| Sub-reviewer                    | 27 <sup>th</sup> ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering (ESEC/FSE) |

### Other Service

|                               |                                                                     |
|-------------------------------|---------------------------------------------------------------------|
| ..... 2018 .....              |                                                                     |
| PhD student representative    | Manning College of Information and Computer Sciences, UMass Amherst |
| ..... 2017 .....              |                                                                     |
| Event-coordinator             | Indian Students Association (ISA) committee, UMass Amherst          |
| ..... 2014 .....              |                                                                     |
| Event-coordinator             | Recreational Activities Organization committee, TRDDC, Pune         |
| ..... 2010 .....              |                                                                     |
| Organization committee member | Felicity, annual cultural and technical fest of IIIT Hyderabad      |

### Professional Associations

|        |                                           |                |
|--------|-------------------------------------------|----------------|
| Member | Association for Computing Machinery (ACM) | 2012 – present |
|--------|-------------------------------------------|----------------|